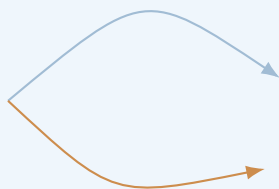


Physics Lectures

Classical Field Theory I

Lecture Notes

Foundations, variational principles, and fundamental classical fields.



June 9, 2026

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Preface

0.1 Classical Field Theory as a Bridge Between Classical Mechanics and Quantum Field Theory

Classical field theory is one of the central languages of modern theoretical physics. It begins from ideas that are already familiar in classical mechanics: configuration, motion, action, symmetry, and conservation. It then replaces a finite list of generalized coordinates by fields, whose degrees of freedom are spread continuously over space and time.

In mechanics, a system may be described by coordinates

$$q^1(t), \dots, q^n(t).$$

In field theory, the dynamical variables are functions such as

$$\phi(t, \mathbf{x}), \quad A_\mu(x), \quad \psi_\alpha(x).$$

The difference is not merely notational. A field has independent values at different points, and its equations usually involve partial derivatives with respect to both time and position. This is why field theory is the natural language of waves, electromagnetism, continua, gauge interactions, and the classical limit of quantum field theory.

These notes are written as a bridge. On one side stands classical mechanics, especially the Lagrangian and Hamiltonian viewpoints. On the other side stand quantum field theory, gauge theory, and relativistic physics. The aim is not to quantize fields in this course, but to make the classical structures precise enough that their later quantization is not mysterious.

Remark

Classical field theory is not a historical leftover. It is the structural language behind many modern theories. Even when a theory is ultimately quantum, its action, symmetries, currents, constraints, and field equations are usually first understood classically.

0.2 What Is a Field? From Particles to Continuous Degrees of Freedom

A particle model assigns a finite amount of data to each particle: position, velocity, momentum, charge, spin, or other variables. A field model assigns data to each point of a continuous domain. For example, a temperature field assigns a number to each point in a body, while an electromagnetic field assigns electric and magnetic vectors to each point of space and time.

The move from particles to fields is a move from finitely many degrees of freedom to continuously many degrees of freedom. This is why field theory is mathematically richer than point-particle mechanics. It involves functions, partial differential equations, boundary conditions, and local conservation laws.

The basic question is always:

What physical quantity is assigned to each point?

For a scalar field, the answer is one number. For a vector field, it is a vector. For a tensor field, it is a multilinear object whose components depend on coordinates but whose meaning should not depend on the chosen coordinate system.

Definition

A *classical field* is a physical quantity represented by an assignment of mathematical data to each point of space, spacetime, or another continuous domain. The value assigned at a point may be a scalar, vector, tensor, spinor, or more general geometric object.

The word *classical* means that the field is treated as a definite object, not as an operator or quantum state. Classical field theory studies equations for such fields before quantization.

0.3 Models, Assumptions, Equations, and Interpretation

An equation is never the whole theory by itself. A field equation becomes a physical model only after one specifies the field, the domain, the assumptions, the interpretation of the variables, and the class of allowed solutions.

For instance, the wave equation

$$\frac{\partial^2 \phi}{\partial t^2} - c^2 \Delta \phi = 0$$

may describe small vibrations of a string, sound waves in an idealized medium, or a simple relativistic scalar field after a change of notation. The same formal equation can therefore represent different physics. The interpretation comes from what ϕ means, what c means, what the domain is, and what approximations led to the equation.

Throughout these notes, a model will be read through four layers:

assumptions $\longrightarrow$ mathematical variables $\longrightarrow$ field equations $\longrightarrow$ physical interpretation.

This order matters. If the assumptions are changed, the same equation may no longer be appropriate. If the interpretation is changed, the same symbols may refer to different observables.

Summary

The goal is not only to manipulate field equations. The goal is to understand what each equation says, what it ignores, which quantities it conserves, and which physical questions it can answer.

0.4 How to Read a Field Equation

A field equation should be read locally first. A typical equation relates the field and its derivatives at a point. For example, an equation of the form

$$\partial_\mu \partial^\mu \phi + m^2 \phi = 0$$

states that the local curvature of the field configuration, in spacetime, is related to the local value of the field itself.

Several questions should be asked whenever a new field equation appears.

1. What is the unknown field?
2. What is the domain: space, spacetime, or a curved manifold?
3. Which derivatives appear, and what do they measure?
4. What constants or parameters occur, and what are their dimensions?

5. What initial or boundary data are needed?
6. Does the equation come from an action principle?
7. Which symmetries and conservation laws does it have?

This habit prevents a common mistake: treating field equations as isolated formulas. In physics, an equation is a compressed statement about a model. It must be unpacked.

Example

Maxwell's equations are not only four equations for $\mathbf{E}$ and $\mathbf{B}$. They also encode locality, charge conservation, gauge structure, propagation of electromagnetic waves, and the coupling between fields and sources.

0.5 Notation, Conventions, Units, and Index Language

These notes use standard mathematical notation for fields and their derivatives. Spatial vectors are written in boldface, for example $\mathbf{x}$ and $\mathbf{E}$. Spacetime indices are usually Greek:

$$\mu, \nu, \rho, \sigma = 0, 1, 2, 3.$$

Spatial indices are usually Latin:

$$i, j, k = 1, 2, 3.$$

Repeated upper and lower indices are summed when index notation is used.

The derivative ∂_μ denotes differentiation with respect to the coordinate x^μ . In relativistic notation, field equations are often written so that their transformation properties under Lorentz transformations are visible. When the metric is needed, the convention will be stated before it is used.

Units are part of the physics. Sometimes constants such as c , $\hbar$, or ε_0 will be displayed explicitly to clarify dimensions. In more theoretical sections, natural units may be used to simplify formulas. The text will indicate when such simplifications are made.

Remark

Index notation is a language, not a decoration. It records which objects are scalars, vectors, covectors, tensors, and densities. A formula with indices should be read together with its transformation meaning.

0.6 Recommended Mathematical Background

The main mathematical prerequisites are classical mechanics, multivariable calculus, linear algebra, and ordinary differential equations. The reader should be comfortable with gradients, divergence, curl, integration by parts, matrices, eigenvectors, and basic variational ideas.

Partial differential equations appear throughout the course. Full mathematical rigour about existence and uniqueness is not the main goal, but the reader should understand that field equations require initial data, boundary data, and conditions on the domain.

Some tensor notation and special relativity will also be used. The necessary material is reviewed inside the notes, but the reader should expect the notation to become more geometric as the course develops. Differential forms, Lie groups, and spinors are introduced only as much as needed for classical field theory.

Summary

The mathematical attitude needed for these notes is more important than a long list of prerequisites: follow definitions carefully, track assumptions, check dimensions, and ask what each symbol represents physically.

0.7 Structure of the Course

The course is organized in three main parts, followed by appendices.

Part I develops the foundations. It explains how field theory grows out of mechanics, introduces the mathematical language of fields, and reviews the spacetime formulation needed for relativistic theories.

Part II develops the Lagrangian and Hamiltonian formalisms for fields. The central objects are the action functional, the Lagrangian density, the Euler–Lagrange equations, symmetries, Noether currents, conserved charges, and canonical phase space.

Part III studies fundamental classical fields: real and complex scalar fields, the electromagnetic field, non-Abelian gauge fields, and spinor fields. These examples are chosen because they form the classical backbone of many later quantum and gauge-theoretic constructions.

The appendices collect conventions and technical tools: variational calculus, Lie groups and Lie algebras, differential forms, Green’s functions, and problem sets. They are meant to support the main text without interrupting the main line of development.

Remark

The book should be read actively. A field equation should be differentiated, varied, dimensionally checked, solved in simple cases, and interpreted. The notes are designed to grow by this kind of work.

Part I

**Foundations of Classical Field
Theory**

Chapter 1

From Mechanics to Fields

Remark

This chapter explains how the finite-dimensional language of classical mechanics becomes the infinite-dimensional language of fields.

- 1.1 Generalized Coordinates in Classical Mechanics
- 1.2 Systems with Finitely Many Degrees of Freedom
- 1.3 The Continuum Limit
- 1.4 Fields as Infinite-Dimensional Dynamical Systems
- 1.5 Examples: String, Membrane, Fluid Density, Electromagnetic Field
- 1.6 Initial Data and Boundary Data
- 1.7 Locality and Causality
- 1.8 What Makes a Theory a Field Theory?

Chapter 2

Mathematical Preliminaries

Remark

This chapter collects the mathematical tools needed before the action principle for fields can be used cleanly.

2.1 Scalar, Vector, Tensor, and Spinor Fields

2.2 Coordinates, Indices, and Summation Convention

2.3 Partial Derivatives and Functional Dependence

2.4 Differential Forms: A Minimal Introduction

2.5 Integration on Space and Spacetime

2.6 Delta Functions and Distributions

2.7 Variations of Fields

2.8 Functional Derivatives

2.9 Boundary Terms and the Meaning of “Up to a Total Derivative”

2.10 Lorentzian Geometry Needed for Field Theory

Chapter 3

Special Relativity and Spacetime Formulation

3.1 Minkowski Spacetime

3.2 Lorentz Transformations

3.3 Four-Vectors and Tensors

3.4 Proper Time, Worldlines, and Causal Structure

3.5 Relativistic Notation for Fields

3.6 Covariant and Non-Covariant Forms of Equations

3.7 Invariant Actions

3.8 Why Relativistic Field Theory Is Natural

Part II

Lagrangian Field Theory

Chapter 4

The Action Principle for Fields

4.1 The Action Functional

4.2 Lagrangian Density

4.3 Variational Principle for Fields

4.4 Euler–Lagrange Equations for Fields

4.5 Boundary Conditions in the Variational Principle

4.6 Total Derivatives and Equivalent Lagrangians

4.7 On-Shell and Off-Shell Statements

4.8 Worked Example: Real Scalar Field

4.9 Worked Example: Wave Equation from an Action

4.10 Physical Interpretation of Field Equations

Chapter 5

Symmetries and Noether's Theorems

5.1 Continuous Symmetries in Field Theory

5.2 Internal and Spacetime Symmetries

5.3 Noether's First Theorem

5.4 Conserved Currents and Conserved Charges

5.5 Energy, Momentum, and Angular Momentum

5.6 Stress-Energy Tensor: Canonical Form

5.7 Improvement Terms

5.8 Noether's Second Theorem and Gauge Symmetries

5.9 Global vs Local Symmetries

5.10 Physical Meaning of Conservation Laws

5.11 Examples and Common Misinterpretations

Chapter 6

Hamiltonian Formulation of Field Theory

6.1 Canonical Momenta for Fields

6.2 Legendre Transform and Hamiltonian Density

6.3 Equal-Time Poisson Brackets

6.4 Hamilton's Equations for Fields

6.5 Phase Space of a Field Theory

6.6 Constraints: First Encounter

6.7 Primary and Secondary Constraints

6.8 Gauge Freedom and Degenerate Legendre Transform

6.9 Energy as a Generator of Time Evolution

6.10 Comparison with the Lagrangian Formulation

Part III

Fundamental Classical Fields

Chapter 7

The Real Scalar Field

7.1 Klein–Gordon Field

7.2 Massless and Massive Scalar Fields

7.3 Plane-Wave Solutions

7.4 Dispersion Relation

7.5 Energy and Momentum of the Scalar Field

7.6 Green’s Functions for the Klein–Gordon Operator

7.7 Sources and Response

7.8 Interacting Scalar Fields

7.9 ϕ^4 Theory as a Classical Model

7.10 Stability of the Potential

7.11 Small Fluctuations Around a Background Solution

Chapter 8

Complex Scalar Field and Internal Symmetry

8.1 Complex Field as Two Real Fields

8.2 Global $U(1)$ Symmetry

8.3 Conserved Current and Charge

8.4 Coupling to an External Electromagnetic Potential

8.5 Local Phase Symmetry

8.6 Minimal Coupling

8.7 Gauge Covariant Derivative

8.8 Physical Meaning of Charge

8.9 Charged Scalar Field as a Classical Field Theory

Chapter 9

Electromagnetic Field

9.1 Maxwell Equations in Vector Form

9.2 Electromagnetic Potentials

9.3 Gauge Freedom

9.4 Maxwell Theory from an Action

9.5 Field Strength Tensor $F_{\mu\nu}$

9.6 Maxwell Equations in Covariant Form

9.7 Coupling to Charged Matter

9.8 Lorenz Gauge and Coulomb Gauge

9.9 Electromagnetic Stress-Energy Tensor

9.10 Energy, Momentum, and Poynting Vector

9.11 Electromagnetic Waves

9.12 Retarded Potentials and Radiation

9.13 Physical Degrees of Freedom of the Electromagnetic Field

Chapter 10

Gauge Fields and Yang–Mills Theory

10.1 From Electromagnetism to Non-Abelian Gauge Fields

10.2 Lie Groups and Lie Algebras: Minimal Toolkit

10.3 Gauge Connection

10.4 Curvature / Field Strength

10.5 Gauge Transformations

10.6 Covariant Derivative

10.7 Yang–Mills Action

10.8 Yang–Mills Equations

10.9 Bianchi Identity

10.10 Self-Interaction of Gauge Fields

10.11 Energy-Momentum Tensor of Yang–Mills Fields

10.12 Classical Meaning of Non-Abelian Charge

10.13 Comparison: Abelian vs Non-Abelian Theory

Chapter 11

Spinor Fields and the Dirac Equation

- 11.1 Why Spinors Are Needed
- 11.2 Clifford Algebra and Gamma Matrices
- 11.3 Lorentz Transformations of Spinors
- 11.4 Dirac Field and Dirac Adjoint
- 11.5 Dirac Lagrangian
- 11.6 Dirac Equation from the Variational Principle
- 11.7 Conserved Dirac Current
- 11.8 Coupling to Electromagnetism
- 11.9 Chiral Decomposition
- 11.10 Weyl and Majorana Spinors
- 11.11 Classical Spinor Fields vs Quantum Fermions
- 11.12 What Is Classical and What Is Already Anticipating QFT?

Appendix A

Notation and Conventions

A.1 Metric Signatures

A.2 Index Conventions

A.3 Units and Dimensions

A.4 Fourier Transform Conventions

A.5 Levi-Civita Symbols and Tensors

A.6 Differential Form Conventions

A.7 Gamma Matrix Conventions

Appendix B

Variational Calculus for Fields

B.1 Variation of a Functional

B.2 Integration by Parts

B.3 Boundary Terms

B.4 Functional Derivatives

B.5 Variation with Constraints

B.6 Variation with Respect to the Metric

B.7 Common Variational Identities

Appendix C

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C.1 Groups, Algebras, and Generators

C.2 Structure Constants

C.3 Representations

C.4 Adjoint Representation

C.5 Exponential Map

C.6 Gauge Transformations

C.7 Examples: $U(1)$, $SU(2)$, $SU(3)$, $SO(3)$, $SO(1,3)$

Appendix D

Differential Forms and Geometry

D.1 Forms and Exterior Derivative

D.2 Wedge Product

D.3 Hodge Dual

D.4 Stokes' Theorem

D.5 Connections and Curvature

D.6 Bianchi Identities

D.7 Applications to Maxwell and Yang–Mills Theory

D.8 Applications to Gravity

Appendix E

Green's Functions

E.1 Green's Functions for Ordinary Differential Equations

E.2 Green's Functions for Partial Differential Equations

E.3 Retarded, Advanced, and Feynman-Type Green's Functions

E.4 Green's Functions for the Wave Equation

E.5 Green's Functions for the Klein–Gordon Equation

E.6 Physical Interpretation as Response to Sources

Appendix F

Suggested Problems

F.1 Variational Principle

F.2 Noether Currents

F.3 Scalar Fields

F.4 Electromagnetism

F.5 Yang–Mills Theory

F.6 Spinor Fields

F.7 Spontaneous Symmetry Breaking

F.8 Solitons

F.9 Gravity as Field Theory

F.10 Boundaries and Charges

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